

Calculate the fair coupon of a USD basket autocallable

Please calculate the **fair annual coupon yyy** such that the following autocallable has an initial fair value of **USD 100 per USD 100 principal**.

Use the supplied inputs below. Clearly distinguish actual calculations or simulations from assumptions. Do not invent missing Monte Carlo results.

1. Basic terms

- **Currency:** USD.
- **Principal:** USD 100.
- **Maximum maturity:** 5 years.
- **Observation frequency:** Annual, at years 1–5.
- **USD risk-free rate: 5.69%**, supplied as a modeling assumption—not a coupon or a request to retrieve today's rate.
- No interest-rate term structure or compounding convention has been supplied. For the base case, assume a flat **annually compounded** rate:
 $D(0,t) = (1.0569)^{-t}$.
Also show sensitivity to interpreting 5.69% as continuously compounded.
- **Unknown annual coupon:** yyy.
- **Coupon accrual:** Simple, $y \times t$, not compounded.
- Assume no interim coupon payments.
- If first autocalled in year kkk, the total redemption is:
 $100(1+ky)$.
- Ignore issuer default, fees, and hedging margins in the base valuation, but identify these exclusions.

2. Basket weights

Use a fixed-weight arithmetic basket of normalized USD asset performances:

$$B_t = \sum_i w_i \frac{S_i^{\text{USD}}(t)}{S_i^{\text{USD}}(0)}, B_0 = 1. \quad B_t = \sum_i w_i \frac{S_i^{\text{USD}}(t)}{S_i^{\text{USD}}(0)}, B_0 = 1.$$

Asset	Ticker	Weight
Asia semiconductor ETF	3119.HK	35%
South Korea ETF	EWY	20%

Taiwan core ETF	EWT	15%
Japan semiconductor ETF	2644.T	15%
Taiwan ESG ETF	00878.TW	15%

All asset performances are **already converted to USD**. Do not apply a second currency conversion.

However, it has not been established whether the supplied volatility and correlation inputs were estimated on USD returns or local-currency returns. Identify this limitation if independently simulating the basket.

3. Autocall schedule

The note terminates at the **first annual observation** when $B_k \geq H_k$ or $H_k \geq B_k$.

Year	Autocall trigger
1	102.5%
2	100.0%
3	97.5%
4	95.0%
5	92.5%

4. Maturity redemption and 70% barrier

If the note never autocalls, the principal redemption at year 5 is:

$$100R(B_5), R(B_5) = \begin{cases} 1, & B_5 \geq 0.70 \\ B_5, & B_5 < 0.70 \end{cases} \quad 100R(B_5), R(B_5) = \begin{cases} 1, & B_5 \geq 0.70 \\ B_5, & B_5 < 0.70 \end{cases}$$

Examples:

- Final basket at 65% → principal redemption of USD 65.
- Final basket at 77% → principal redemption of USD 100.

Treat this as a **maturity-only downside barrier**, not a guaranteed 70% minimum repayment.

The coupon treatment on never-called paths remains unspecified. Please calculate both:

- **Case A:** No coupon on never-called paths.
- **Case B:** Never-called paths receive an additional 100(5y)100(5y)100(5y), regardless of final basket level, on top of the principal redemption.

Apply the year-5 autocall test before applying the never-called maturity payoff.

5. Implied volatility term structures

Annualized volatilities, expressed as decimals:

Ticker	1Y	2Y	3Y	4Y	5Y
3119.HK	0.3045	0.3234	0.335	0.367	0.375
EWY	0.415	0.427	0.445	0.445	0.445
EWT	0.324	0.358	0.363	0.363	0.363
2644.T	0.230	0.249	0.256	0.256	0.259
00878.TW	0.070	0.090	0.120	0.120	0.120

Data caveat: The Japanese volatility row was initially labeled “2664,” while the subsequently supplied basket ticker was “2644 JP.” Provisionally assign that row to 2644.T, but flag this as needing confirmation.

For a deterministic-volatility lognormal model, derive interval variances from total variance:

$$\Delta V_{i,k} = t_k \sigma_i(t_k)^2 - t_{k-1} \sigma_i(t_{k-1})^2$$

$$= t_k \sigma_i(t_k)^2 - t_{k-1} \sigma_i(t_{k-1})^2$$

Do not directly treat maturity-specific implied volatilities as the corresponding annual interval volatilities.

6. Correlation matrix

Use the following supplied correlations:

	3119.HK	EWY	EWT	2644.T	00878.TW
3119.HK	1	0.534	0.741	0.863	0.562
EWY	0.534	1	0.587	0.491	0.438
EWT	0.741	0.587	1	0.614	0.689
2644.T	0.863	0.491	0.614	1	0.517
00878.TW	0.562	0.438	0.689	0.517	1

Check that the matrix is positive semidefinite before simulation.

7. Existing risk-neutral Monte Carlo results

The supplied **cumulative probabilities of having autocalled by each year** are:

Year	Cumulative call probability	Implied first-call probability
1	78%	78%
2	93%	15%
3	97%	4%
4	99%	2%
5	99%	0%

Thus, **1% never autocalls**.

These have been confirmed as risk-neutral probabilities. Treat them as supplied, rounded estimates—not independently verified results. The displayed 0% year-5 first-call probability may reflect rounding.

8. Requested analysis

A. Value directly from the supplied probabilities

For Case A, use:

$$1 = \sum_{k=1}^5 p_k D_k (1+ky) + D_5 EQ[1\{\text{never called}\}R(B_5)].$$

$$1 = \sum_{k=1}^5 p_k D_k (1+ky) + D_5 E^{\mathbb{Q}} \left[\mathbb{1}_{\{\text{never called}\}} R(B_5) \right].$$

Therefore:

$$y_{\text{fair}} = \frac{1 - \sum_{k=1}^5 p_k D_k - D_5 E^{\mathbb{Q}} [\mathbb{1}_{\{\text{never called}\}} R(B_5)]}{\sum_{k=1}^5 p_k D_k}.$$

For Case B, add $D_5 P(\text{never called})$ to the denominator.

The conditional maturity redemption distribution has **not** been supplied. Do not assume that a 70% barrier means an average repayment of 70%.

Provide:

- Fair-coupon sensitivity to average never-called repayment fractions of **100%, 90%, 80%, 70%, 60%, and 0%**.
- Upper and lower fair-coupon bounds using the supplied rounded probabilities.
- The exact simulation statistic needed to obtain a single fair coupon.

B. Independently check the simulation, if possible

Using the weights, volatility term structures, correlations, and 5.69% rate:

- Explain any additional dividend, forward, or USD-return assumptions required.
- If you run Monte Carlo, report the model, path count, random seed, and statistical uncertainty.
- Compare simulated call probabilities with the supplied CDF, especially the **78% first-year call probability at a 102.5% trigger**.
- Do not force agreement or combine a new model's tail-loss estimate with the supplied CDF without explaining the consistency problem.
- If code execution is unavailable, provide reproducible Python code rather than claiming to have run a simulation.

End with the defensible fair-coupon estimate or range, its assumptions, and the remaining information required for an exact valuation.
